

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure A

Experiments

Criteria	Understanding of Concepts (2.5)	Experimental Design (3)	Use of Tools (2)	Analysis of Results (1.5)	Documentation (1)	Total(10)
Weightage	25%	30%	20%	15%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

*I **Manibaalakrishnan S (Reg.no.:192211171)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Manibaalakrishnan S

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Annexure C

Experiments :

Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

Age of child	Photograph:		
	A	B	C
5-6 years:	18	22	20
7-8 years:	2	28	40
9-10 years:	20	10	40

Use `cov()` to calculate the sample covariance between B and C.

Use another call to `cov()` to calculate the sample covariance matrix for the preferences.

Use `cor()` to calculate the sample correlation between B and C.

Use another call to `cor()` to calculate the sample correlation matrix for the preferences.

Output:

```
--- Sample Covariance between B and C ---
> cov_BC <- cov(B, C)
> cat("cov(B, C) =", cov_BC, "\n")
cov(B, C) = -20
>
> cat("\n--- Sample Covariance Matrix (A, B, C) ---\n")

--- Sample Covariance Matrix (A, B, C) ---
> cov_matrix <- cov(photo_data)
> print(cov_matrix)
      A      B      C
A  97.33333 -74 -46.66667
B -74.00000  84 -20.00000
C -46.66667 -20 133.33333
>
> # ---- Correlation ----
> cat("\n--- Sample Correlation between B and C ---\n")

--- Sample Correlation between B and C ---
> cor_BC <- cor(B, C)
> cat("cor(B, C) =", cor_BC, "\n")
cor(B, C) = -0.1889822
>
> cat("\n--- Sample Correlation Matrix (A, B, C) ---\n")

--- Sample Correlation Matrix (A, B, C) ---
> cor_matrix <- cor(photo_data)
> print(cor_matrix)
      A      B      C
A  1.0000000 -0.8183918 -0.4096440
B -0.8183918  1.0000000 -0.1889822
C -0.4096440 -0.1889822  1.0000000
```

```

--- Part (a): Descriptive Statistics ---
> stats_summary <- data.frame(
+   Metric = c("Mean", "Median", "Std Dev"),
+   Age     = c(round(mean(age), 2), median(age), round(sd(age), 2)),
+   PercentFat = c(round(mean(fat), 2), median(fat), round(sd(fat), 2))
+ )
> print(stats_summary)
  Metric   Age PercentFat
1   Mean 46.44      28.78
2  Median 51.00      30.70
3 Std Dev 13.22       9.25

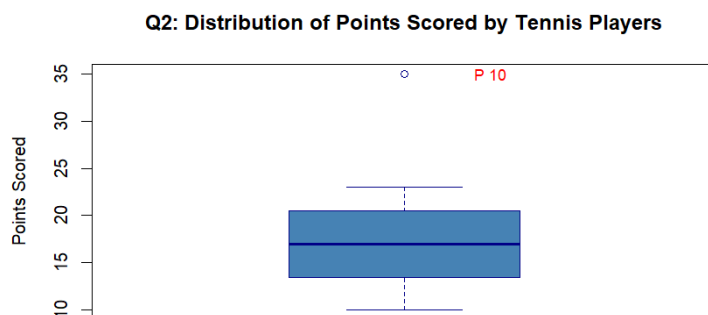
```

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player ID Player Name Points Scored

P1	Player A	12
P2	Player B	15
P3	Player C	14
P4	Player D	10
P5	Player E	18
P6	Player F	20
P7	Player G	22
P8	Player H	13
P9	Player I	11
P10	Player J	35
P11	Player K	16
P12	Player L	17
P13	Player M	19
P14	Player N	21
P15	Player O	23

Output:

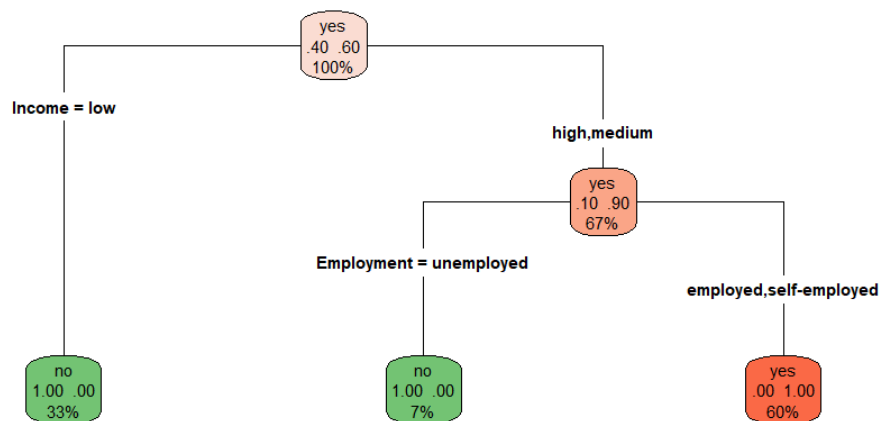


3. A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Output:

Q3: Loan Approval Decision Tree



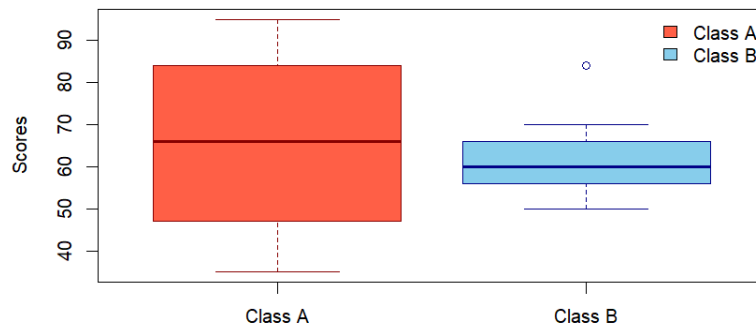
Using historical data, build a **Decision Tree classifier** to predict **Loan Approval (Yes/No)**.
(10 Marks)

Consider the Data Set

young,high,employed,good,yes
 young,high,self-employed,average,yes
 middle,high,employed,good,yes
 old,medium,employed,good,yes
 old,low,unemployed,poor,no
 old,low,self-employed,average,no
 middle,low,unemployed,poor,no
 young,medium,employed,average,yes
 young,low,unemployed,poor,no
 old,medium,self-employed,good,yes
 middle,medium,employed,average,yes
 middle,high,self-employed,good,yes
 old,medium,unemployed,poor,no
 young,medium,self-employed,average,yes
 middle,low,employed,poor,no

Output:

Q4: Year 9 Maths Exam Scores - Class A vs Class B



4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 84, 76, 35, 47, 64, 95, 66, 89, 36, 84

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 50, 51, 56, 84, 60, 59, 70, 63, 66, 50

(i) Find which class had scored higher mean, median and range.

(ii) Plot above in boxplot and give the inferences

Output:

```
--- Part (i): Descriptive Statistics ---
> stats_df <- data.frame(
+   Statistic = c("Mean", "Median", "Range"),
+   Class_A   = c(mean_A, median_A, range_A),
+   Class_B   = c(mean_B, median_B, range_B)
+ )
> print(stats_df)
  Statistic Class_A Class_B
1     Mean 65.7778 62.1111
2    Median 66.00000 60.00000
3     Range 60.00000 34.00000
>
> cat("\nConclusion:\n")

Conclusion:
> cat("Higher Mean: ", ifelse(mean_A > mean_B, "Class A", "Class B"),
+   " (A = ", round(mean_A,2), ", B = ", round(mean_B,2), ") \n")
Higher Mean: Class A (A = 65.78 , B = 62.11 )
> cat("Higher Median: ", ifelse(median_A > median_B, "Class A", "Class B"),
+   " (A = ", median_A, ", B = ", median_B, ") \n")
Higher Median: Class A (A = 66 , B = 60 )
> cat("Larger Range: ", ifelse(range_A > range_B, "Class A (more spread out)",
+   "Class B (more spread out)"),
+   " (A = ", range_A, ", B = ", range_B, ") \n")
Larger Range: Class A (more spread out) (A = 60 , B = 34 )
>
> # Part (ii) - Boxplot
> boxplot(class_A, class_B,
+   names = c("Class A", "Class B"),
+   col = c("tomato", "skyblue"),
+   border = c("darkred", "darkblue"),

--- Part (ii): Boxplot Inferences ---
> cat("1. Class A has a higher median and mean, so Class A performed better overall.\n")
1. Class A has a higher median and mean, so Class A performed better overall.
> cat("2. Class A has a much larger range (", range_A, ") vs Class B (",
+   range_B, ") - more variable performance.\n")
2. Class A has a much larger range ( 60 ) vs Class B ( 34 ) - more variable performance.
> cat("3. Class B scores are more tightly clustered (consistent performance).\n")
3. Class B scores are more tightly clustered (consistent performance).
> cat("4. Class A shows potential outliers at the low end (score: 35).\n")
4. Class A shows potential outliers at the low end (score: 35).
> cat("5. Both classes have at least one high scorer (84-95 range).\n")
5. Both classes have at least one high scorer (84-95 range).
>
```

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

<i>age</i>	23	23	27	27	39	41	47	49	50
<i>%fat</i>	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2

<i>age</i>	52	54	54	56	57	58	58	60	61
<i>%fat</i>	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

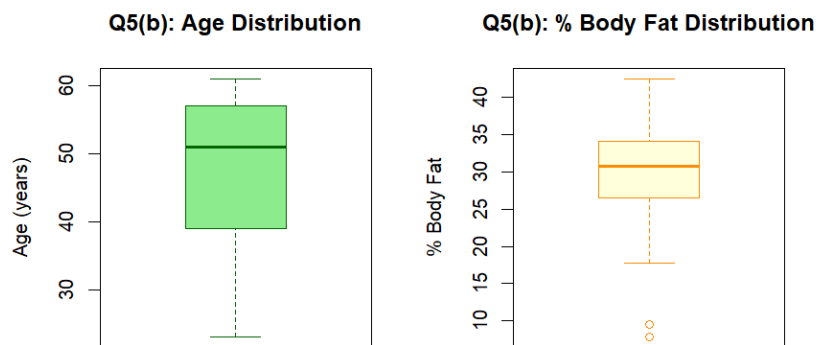
- Calculate the mean, median, and standard deviation of *age* and *%fat*.
- Draw the boxplots for *age* and *%fat*.
- Draw a *scatter plot* and a *q-q plot* based on these two variables.

Output:

```

--- Part (a): Descriptive Statistics ---
> stats_summary <- data.frame(
+   Metric = c("Mean", "Median", "Std Dev"),
+   Age     = c(round(mean(age), 2), median(age), round(sd(age), 2)),
+   PercentFat = c(round(mean(fat), 2), median(fat), round(sd(fat), 2))
+ )
> print(stats_summary)
  Metric   Age PercentFat
1   Mean 46.44      28.78
2  Median 51.00      30.70
3 Std Dev 13.22       9.25

```



```

--- Part (c): Scatter Plot ---
> plot(age, fat,
+   main = "Q5(c): Age vs % Body Fat",
+   xlab = "Age (years)",
+   ylab = "% Body Fat",
+   pch = 16,
+   col = "steelblue",
+   cex = 1.3)
> abline(lm(fat ~ age), col = "red", lwd = 2, lty = 2)
> legend("topleft", legend = "Regression Line", col = "red",
+   lty = 2, lwd = 2, bty = "n")
>
> cat("Scatter plot shows a positive trend: body fat increases with age.\n")
Scatter plot shows a positive trend: body fat increases with age.
> cat("Correlation:", round(cor(age, fat), 4), "\n")
Correlation: 0.8176

```